

**A PROJECT REPORT ON**  
**“COMPARATIVE STUDY OF SOBEL VS. CANNY FOR**  
**MEDICAL X-RAY BOUNDARY DETECTION”**

Submitted by

**JANDHYALA ADITYA TARUN ,**  
**VANAM DIVESH REDDY,**  
**RANI YADAV.**

23STUCHH010189, 010247, 010308

**Pursuing**

**BACHELOR OF TECHNOLOGY**

**IN**

**DATA SCIENCE & ARTIFICIAL INTELLIGENCE**



**ICFAI Foundation for Higher Education**

Faculty of science and technology, Donthanpally, Shankarpalli Road,  
Hyderabad – 501203, Telangana, India.

**SEPTEMBER-2026**



**ICFAI Foundation for Higher  
Education**

**Faculty of Science and Technology**

Donthanpally, Shankarpalli Road, Hyderabad  
- 501203, Telangana, India.

## **CERTIFICATE**

This is to certify that the project entitled “**COMPARATIVE STUDY OF SOBEL VS. CANNY FOR MEDICAL X-RAY BOUNDARY DETECTION**” by JANDHYALA ADITYA TARUN, VANAM DIVESH REDDY, RANI YADAV (23STUCHH010189, 010247, 010308) has been submitted in the partial fulfilment of the requirements for the award of degree of Bachelor of technology in computer Science and Engineering from Faculty of Science & Technology (FST), IFHE Hyderabad. The results embodied in this project have not been submitted to any other University or Institution for the award of any degree or diploma.

**Head of Department**

Department of AI & Data Science  
IFHE, Hyderabad

**Dr. N. Suresh**

Faculty Mentor  
Department of AI & Data Science  
IFHE, Hyderabad

## ABSTRACT

In medical image analysis, edge detection remains a crucial step for automated systems to distinguish anatomical structures from surrounding tissue based on localized intensity changes. This project compares two classical edge-detection methods, the Sobel operator and the Canny edge detector applied to medical X-ray images to evaluate their effectiveness in boundary detection.

A small, curated set of X-ray images serves as the common input, with identical preprocessing (grayscale conversion and noise reduction) applied to ensure a fair, controlled comparison. The Sobel operator uses horizontal and vertical derivative kernels to estimate gradient magnitude at each pixel, while the Canny detector employs OpenCV's multi-stage pipeline involving Gaussian smoothing, gradient computation, non-maximum suppression, and hysteresis thresholding, with all threshold parameters documented for reproducibility.

The resulting edge maps are assessed qualitatively by visually inspecting edge continuity, thickness, and noise sensitivity, and quantitatively through edge density and processing time metrics. The implementation uses Python with OpenCV, NumPy, and Matplotlib.

Upon completion, this study summarizes which method produced more continuous or thinner boundaries, which was more noise-robust, and which was computationally faster based on measured results. The findings provide evidence-based insights into the trade-offs between gradient-based and multi-stage edge-detection approaches in medical imaging contexts.

| <b>Table of Contents</b> |                                     |                 |
|--------------------------|-------------------------------------|-----------------|
| <b>S. No.</b>            | <b>Contents</b>                     | <b>Page No.</b> |
| <b>1</b>                 | <b>Introduction</b>                 | <b>5</b>        |
| <b>1.1</b>               | <b>Overview</b>                     | <b>5</b>        |
| <b>2</b>                 | <b>Relevance of the Project</b>     | <b>6</b>        |
| <b>2.1</b>               | <b>Historical Evolution</b>         | <b>6</b>        |
| <b>3</b>                 | <b>Methodology</b>                  | <b>8</b>        |
| <b>3.1</b>               | <b>Dataset Description</b>          | <b>8</b>        |
| <b>3.2</b>               | <b>Pipeline</b>                     | <b>8</b>        |
| <b>3.3</b>               | <b>Preprocessing</b>                | <b>10</b>       |
| <b>3.4</b>               | <b>Sobel Edge Detection</b>         | <b>10</b>       |
| <b>3.4.1</b>             | <b>Formulation and kernels</b>      | <b>10</b>       |
| <b>3.4.2</b>             | <b>Implementation</b>               | <b>11</b>       |
| <b>3.5</b>               | <b>Canny Edge Detection</b>         | <b>11</b>       |
| <b>3.5.1</b>             | <b>Formulas and Kernels</b>         | <b>11</b>       |
| <b>3.5.2</b>             | <b>Implementation</b>               | <b>12</b>       |
| <b>4</b>                 | <b>Results and Discussion</b>       | <b>13</b>       |
| <b>4.1</b>               | <b>Output Comparison</b>            | <b>13</b>       |
| <b>4.2</b>               | <b>Code Snippet</b>                 | <b>13</b>       |
| <b>4.3</b>               | <b>Observations</b>                 | <b>16</b>       |
| <b>5</b>                 | <b>Conclusions and Future Scope</b> | <b>18</b>       |
| <b>5.1</b>               | <b>Summary and Key Takeaways</b>    | <b>18</b>       |
| <b>5.2</b>               | <b>Limitations and Future Scope</b> | <b>18</b>       |
| <b>6</b>                 | <b>References</b>                   | <b>19</b>       |

# Chapter 1: Introduction

## 1.1 Overview

This project compares two classical edge-detection techniques the Sobel operator and the Canny edge detector applied to medical X-ray images using computer vision methods. In X-rays, anatomical structures such as bones and soft-tissue boundaries appear as pixel intensity variations rather than sharp outlines. Edge detection transforms these intensity changes into meaningful representations of structural boundaries, which is essential for tasks like computer-aided diagnosis and image segmentation.

The study uses a common processing pipeline on a small set of X-ray images: each image is converted to grayscale and smoothed to reduce noise, then processed independently by the Sobel and Canny algorithms under identical conditions. Sobel detects edges by estimating horizontal and vertical intensity gradients, while Canny employs a multi-stage process including smoothing, gradient calculation, non-maximum suppression, and hysteresis thresholding, to produce thinner, more refined edge maps. Consistent input and preprocessing ensure the comparison isolates differences arising solely from the algorithms.

The outputs are compared both visually (edge continuity, thickness, noise sensitivity) and quantitatively (edge density, processing time), with results summarized in tables and figures. The goal is not to declare a predetermined “winner” but to draw evidence-based conclusions about the suitability of each method for boundary detection in medical X-ray images.

## Chapter 2: Relevance of the project

### 2.1 Historical Evolution

Given below is the history of how this came into existence

#### 1963 - Roberts Cross (Lawrence Roberts)

One of the first edge operators used a pair of  $2 \times 2$  diagonal kernels to approximate the gradient, enabling fast detection of sharp intensity changes. Extremely sensitive to noise due to its tiny kernel size, poor localization on diagonal and curved edges.

#### 1968 to 1970 - Sobel Operator (Irwin Sobel & Gary Feldman)

Introduced  $3 \times 3$  convolution kernels for horizontal and vertical gradients, giving better noise suppression than Roberts through implicit smoothing. Still produces relatively thick, blurred edges; no thinning or edge-linking step, so weak but genuine edges are easily lost or merged with noise

#### 1970 - Prewitt Operator (Judith Prewitt)

Similar  $3 \times 3$  gradient kernel design to Sobel but with uniform weighting, simplifying computation

Slightly less accurate gradient estimation than Sobel due to the lack of weighted emphasis on the center pixels; same thick-edge and noise-sensitivity issues

#### 1980 - Marr–Hildreth / Laplacian of Gaussian (LoG)

Combined Gaussian smoothing with a second-derivative (Laplacian) operator, detecting edges at zero-crossings; grounded in a theory of biological vision

Highly sensitive to the choice of Gaussian scale; can miss edge orientation information and is prone to detecting spurious zero-crossings in textured regions

#### 1986 - Canny Edge Detector (John Canny)

Formalized edge detection as an optimization problem balancing noise suppression, localization, and single-response criteria; combined Gaussian smoothing, gradient computation, non-maximum suppression, and hysteresis thresholding into one pipeline

Computationally more expensive than first-order operators; performance depends heavily on the choice of high/low threshold values and Gaussian sigma, which often require manual tuning

## **1990s to 2000s - Refinements & variants (adaptive thresholding, multi-scale Canny)**

Attempted to automate threshold selection and improve robustness across varying image conditions.

Added complexity without fully resolving parameter sensitivity, still fundamentally limited by hand crafted gradient assumptions

## **2023 - Present**

The research by Pan, J., and others in their 2025 survey shows a major change in how we predict performance. They stress that using Artificial Intelligence (AI) in these models is a huge leap past older, traditional statistical methods. This is mainly because AI can understand and include complex, non-numerical factors like a student's emotions, mindset, and other subtle things that truly affect how well they do in school. By using AI, the models go beyond simple numbers to build a more complete and detailed picture of a student's potential.

## Chapter 3: Methodology

This section outlines the methodology followed to carry out the comparative study of Sobel and Canny edge detection on medical X-ray images. It describes the dataset used, the overall processing pipeline, and the preprocessing steps applied consistently to every image before edge detection. It then details the theoretical basis and implementation specifics of the Sobel and Canny algorithms, the metrics used to compare their outputs, and the tools and project structure used to carry out the implementation. Keeping every stage of the pipeline consistent across both methods ensures that the comparison reflects genuine differences between the algorithms rather than differences in input handling

### 3.1 Dataset Description

The Chest X-Ray Images (Pneumonia) dataset consists of 5,856 validated pediatric chest X-ray images sourced from Guangzhou Women and Children's Medical Center.

#### 1. Structure & Distribution

The dataset is partitioned into three directory splits: train, test, and val.

- **Training Set:** Contains 5,216 total images.
  - **NORMAL:** 1,341 images
  - **PNEUMONIA:** 3,875 images
- **Validation & Test Sets:** Comprise the remaining images for testing model generalizability.

#### 2. Experimental Subset

For the comparative experiment, a balanced subset was sampled from the training directory:

- **Total Sample Size:** 100 images
- **Category Breakdown:** 50 NORMAL images and 50 PNEUMONIA images
- **Sampling Method:** Selected using a fixed random seed to ensure reproducibility.

### 3.2 Pipeline

The edge detection framework evaluates medical chest X-ray images through a standardized, sequential pipeline. This workflow ensures consistency across input samples, removing extraneous background variations before evaluating the image using classical edge detection algorithms

The preprocessing pipeline processes an input chest X-ray image through the following steps:

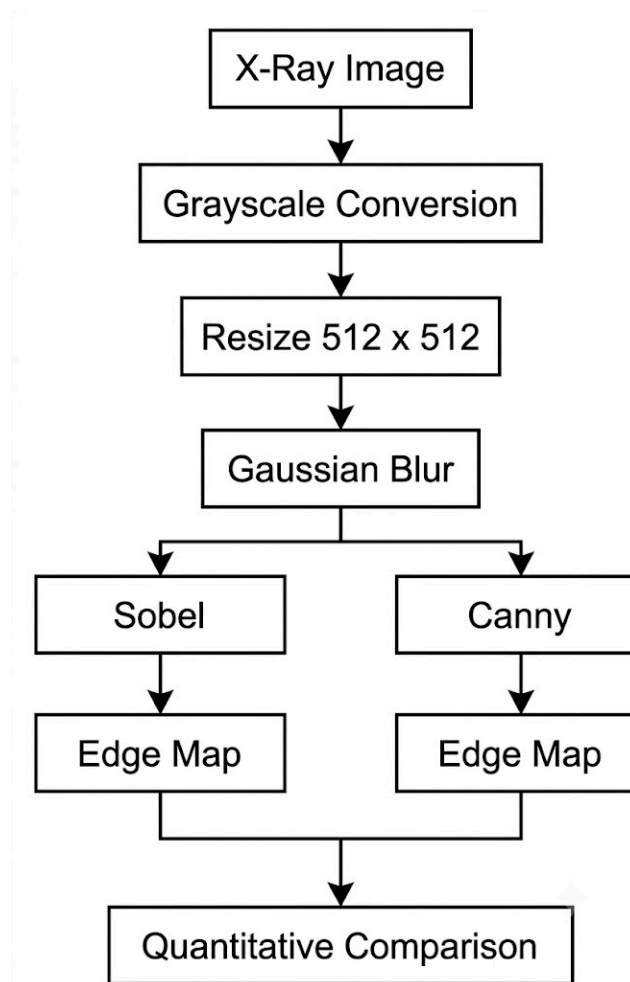
1. **Grayscale Conversion:** Converts the image into a single-channel intensity image suitable for edge analysis.
2. **Resizing:** Standardizes the image dimensions to 512x512 pixels.
3. **Gaussian Blur:** Applies a 5x5 Gaussian kernel filter to reduce small-scale background

noise.

4. **Edge Detection:** Evaluates the preprocessed image using two parallel methods:

- **Sobel:** Computes gradient magnitudes in X and Y directions to generate a Sobel edge map.
- **Canny:** Applies dual-thresholding (50/150) to produce a binary Canny edge map.

5. **Quantitative Comparison:** Compares both generated edge maps based on edge density and execution processing time.



**Figure 1: Project Pipeline**

### 3.3 Preprocessing

To ensure consistent feature extraction and eliminate extraneous noise, all chest X-ray images undergo a uniform three-step preprocessing workflow before applying the edge detection operators:

1. **Grayscale Conversion:** Input images are converted from multi-channel formats into 8-bit single-channel grayscale representations using OpenCV. This reduces memory overhead and isolates intensity luminance values required for computing gradient boundaries.
2. **Spatial Resizing:** Every sample is standardly resized to a uniform spatial resolution of  $512 \times 512$  pixels. Standardizing spatial dimensions normalizes pixel density, making downstream edge pixel counts and processing speed comparisons mathematically consistent across all test cases.
3. **Gaussian Blur Filtering:** A  $5 \times 5$  Gaussian kernel filter is applied across the resized image matrix:

*blurred = cv2.GaussianBlur(resized, (5, 5), 0)*

This low-pass smoothing step suppresses high-frequency background noise and minor imaging artifacts, ensuring that downstream edge detection algorithms capture structural anatomical contours (such as ribs and tissue boundaries) rather than false background noise.

### 3.4 Sobel Edge Detection

The Sobel operator is a spatial gradient-based technique used to emphasize high spatial frequency regions corresponding to boundaries. It calculates the discrete directional derivatives of pixel intensity to approximate the first-order gradient vector across the image.

#### 3.4.1 Formulation and kernels

The algorithm applies two  $3 \times 3$  convolution kernels to compute horizontal ( $G_x$ ) and vertical ( $G_y$ ) intensity changes:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

For every pixel, the total gradient magnitude  $G$  is derived by combining spatial derivatives:

$$G = \sqrt{G_x^2 + G_y^2}$$

### 3.4.2 Implementation

Using OpenCV, directional gradients are extracted using 64-bit floating point precision (cv2.CV\_64F) to avoid overflow artifacts before combining magnitudes:

```
sobel_x = cv2.Sobel(blurred, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(blurred, cv2.CV_64F, 0, 1, ksize=3)
sobel_mag = cv2.magnitude(sobel_x.astype(np.float32), sobel_y.astype(np.float32))
_, sobel_binary = cv2.threshold(sobel_mag, 50, 255, cv2.THRESH_BINARY)
```

A fixed global threshold of **50** is applied to convert continuous magnitude values into a binary edge map for density evaluation.

## 3.5 Canny Edge Detection

The Canny edge detection algorithm is a multi-stage approach designed to optimize edge detection by satisfying three criteria: low error rate, precise localization of edge points, and a single response to a single edge.

### 3.5.1 Formulas and Kernels

Following Gaussian smoothing, the Canny algorithm computes spatial gradient magnitudes ( $G$ ) and angles ( $\Theta$ ) using Sobel derivative operators:

$$G = \sqrt{G_x^2 + G_y^2}, \Theta = \arctan\left(\frac{G_y}{G_x}\right)$$

The algorithm then executes two specialized refinement steps:

1. **Non-Maximum Suppression:** Thinning process that suppresses non-peak gradient magnitudes perpendicular to the edge direction.
2. **Hysteresis Thresholding:** Dual-thresholding mechanism using lower ( $T_{low}$ ) and upper ( $T_{high}$ ) limits to classify pixels:

$$\text{Edge Pixel} = \begin{cases} \text{Strong,} & G \geq T_{high} \\ \text{Weak,} & T_{low} \leq G < T_{high} \text{ (connected to strong)} \\ \text{Non-edge,} & G < T_{low} \end{cases}$$

### 3.5.2 Implementation

Using OpenCV, the Canny pipeline is executed with double-threshold values set to 50 ( $T_{low}$ ) and 150 ( $T_{high}$ ):

```
canny_edges = cv2.Canny(blurred, threshold1=50, threshold2=150)
```

The resulting binary output maps thin, continuous anatomical contours while suppressing background gradient noise.

## Chapter 4: Conclusion and Future Scope

### 4.1 Output Comparison

The visual and qualitative analysis reveals distinct operational differences between the Sobel and Canny edge detection algorithms when applied to preprocessed chest X-ray images:

- **Sobel Output Characteristics:**
  - **Thick, Continuous Gradients:** The Sobel operator computes directional pixel intensity gradients, producing broader, continuous edge regions around primary anatomical structures such as the rib cage, heart border, and clavicles.
  - **Gradient Sensitivity:** Because it uses a simple global magnitude threshold, Sobel captures soft tissue transitions alongside structural boundaries, contributing to a higher average edge density (9.480%).
- **Canny Output Characteristics:**
  - **Thin, Suppressed Edge Maps:** Canny applies non-maximum suppression and hysteresis thresholding, producing fine, single-pixel-wide binary contours while discarding weak, isolated noise.
  - **Isolated Boundary Detail:** The dual-threshold mechanism (50/150) isolates sharp anatomical boundaries while suppressing soft-tissue variance, resulting in a cleaner output with a lower average edge density (**1.873%**).
- **Comparative Observation:**
  - While Sobel highlights broader areas of intensity change across anatomical regions, Canny offers superior structural boundary localization, making its output less cluttered and more defined for feature extraction.

### 4.2 Code Snippet

#### 1. Image resizing and Gaussian smoothing

```
resized = cv2.resize(image, (512, 512))  
  
blurred = cv2.GaussianBlur(resized, (5, 5), 0)
```

#### 2. Sobel gradient calculation and magnitude generation

```
sobel_x = cv2.Sobel(blurred, cv2.CV_64F, 1, 0, ksize=3)  
sobel_y = cv2.Sobel(blurred, cv2.CV_64F, 0, 1, ksize=3)
```

```

sobel_x_abs = cv2.convertScaleAbs(sobel_x)
sobel_y_abs = cv2.convertScaleAbs(sobel_y)

sobel_magnitude = cv2.magnitude(
    sobel_x.astype(np.float32),
    sobel_y.astype(np.float32)
)

sobel_magnitude = cv2.convertScaleAbs(sobel_magnitude)

```

### 3. Canny edge detection using double thresholding

```

canny_edges = cv2.Canny(blurred, threshold1=50, threshold2=150)

```

### 4. Main experimental pipeline for comparing Sobel and Canny across 100 X-ray images for filename, category, folder in image\_samples:

```

image = cv2.imread(
    os.path.join(folder, filename),
    cv2.IMREAD_GRAYSCALE
)

# Resize and smooth
resized = cv2.resize(image, (512, 512))
blurred = cv2.GaussianBlur(resized, (5, 5), 0)

# Sobel
sobel_x = cv2.Sobel(blurred, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(blurred, cv2.CV_64F, 0, 1, ksize=3)

sobel_magnitude = cv2.magnitude(
    sobel_x.astype(np.float32),
    sobel_y.astype(np.float32)
)

# Canny
canny_edges = cv2.Canny(
    blurred,

```

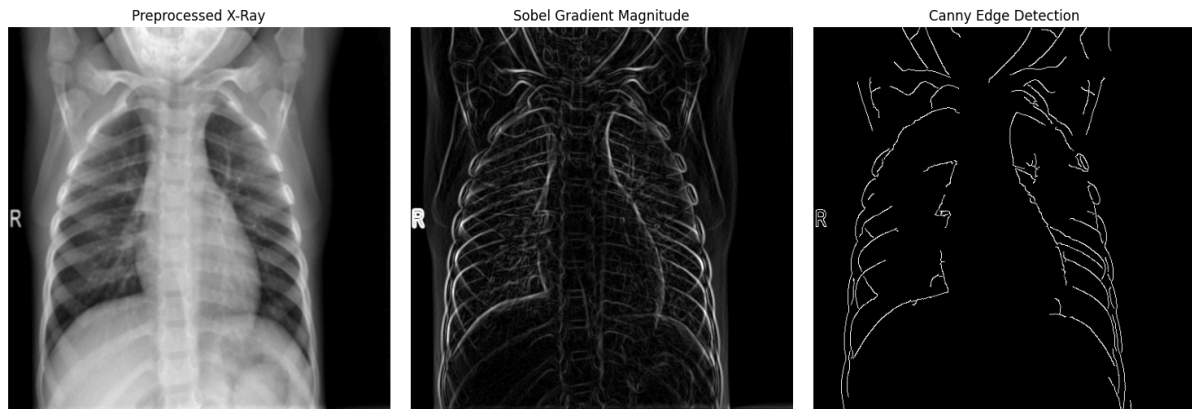
```
threshold1=50,  
threshold2=150  
)
```

## 5. Edge density and processing-time calculation

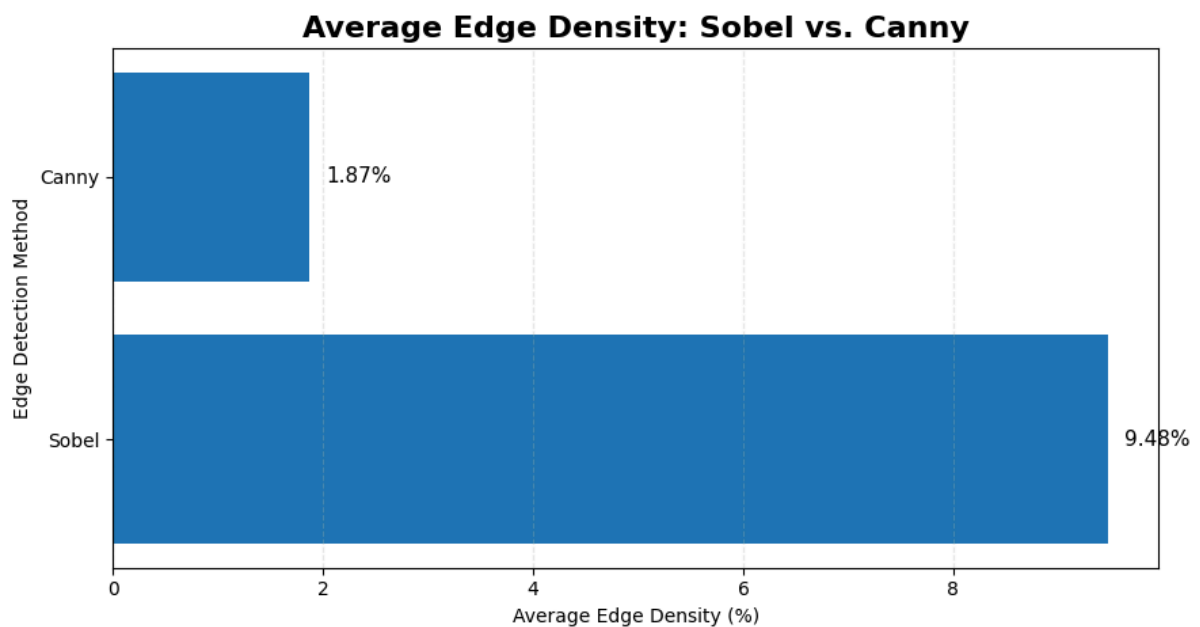
```
sobel_edge_pixels = np.count_nonzero(sobel_binary)  
canny_edge_pixels = np.count_nonzero(canny_edges)  
  
total_pixels = blurred.shape[0] * blurred.shape[1]  
  
# Calculate edge density  
sobel_edge_density = (sobel_edge_pixels / total_pixels) * 100  
canny_edge_density = (canny_edge_pixels / total_pixels) * 100
```

### 4.3 Observations

The experimental results show a clear difference in the edge maps produced by Sobel and Canny for the selected chest X-ray images. Sobel generated denser edge responses across anatomical structures, while Canny produced thinner and more selective boundaries. Across the 100-image dataset, the average edge density was 9.48% for Sobel compared with 1.87% for Canny. The category-wise results followed the same pattern, with Normal images showing higher edge density than Pneumonia images for both methods. The measured average processing time was 2.873 ms for Sobel and 1.108 ms for Canny in the final plotted experiment. The threshold experiment further showed that changing Canny's threshold values altered the resulting edge representation. Overall, the execution demonstrated that the two methods produced noticeably different boundary representations and computational measurements under the selected experimental conditions.



**Figure 2: Visual comparison of the preprocessed X-ray, Sobel gradient magnitude, and Canny edge detection outputs.**



**Figure 3: Comparison of average edge density obtained using Sobel and Canny across**

the 100 selected X-ray images.

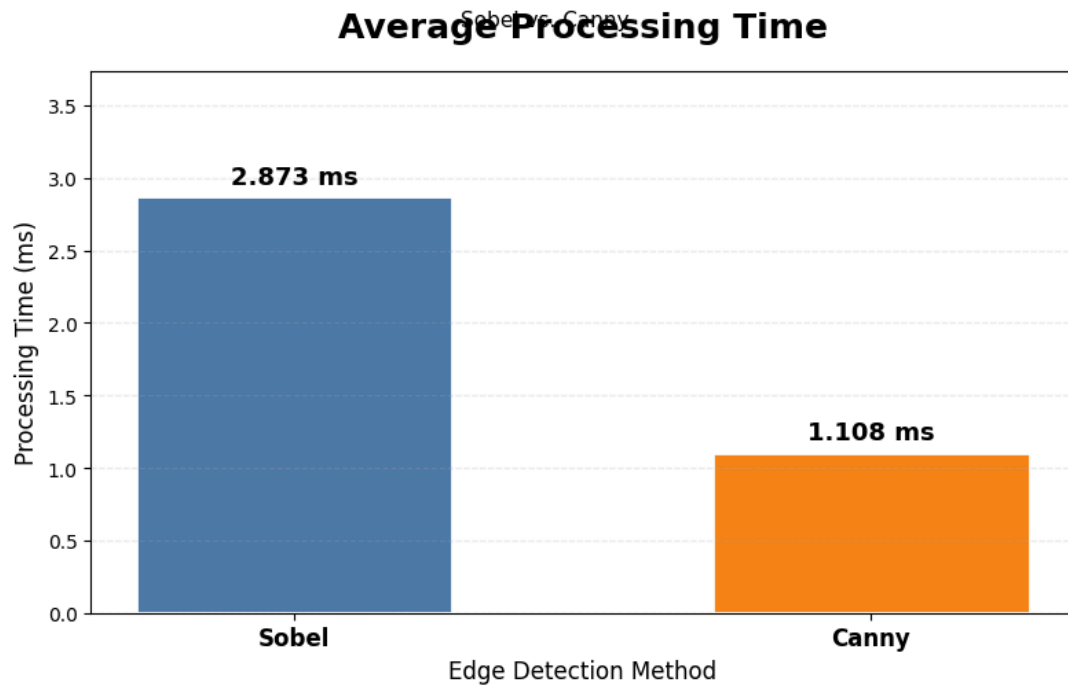


Figure 4: Average processing time of Sobel and Canny edge detection methods.

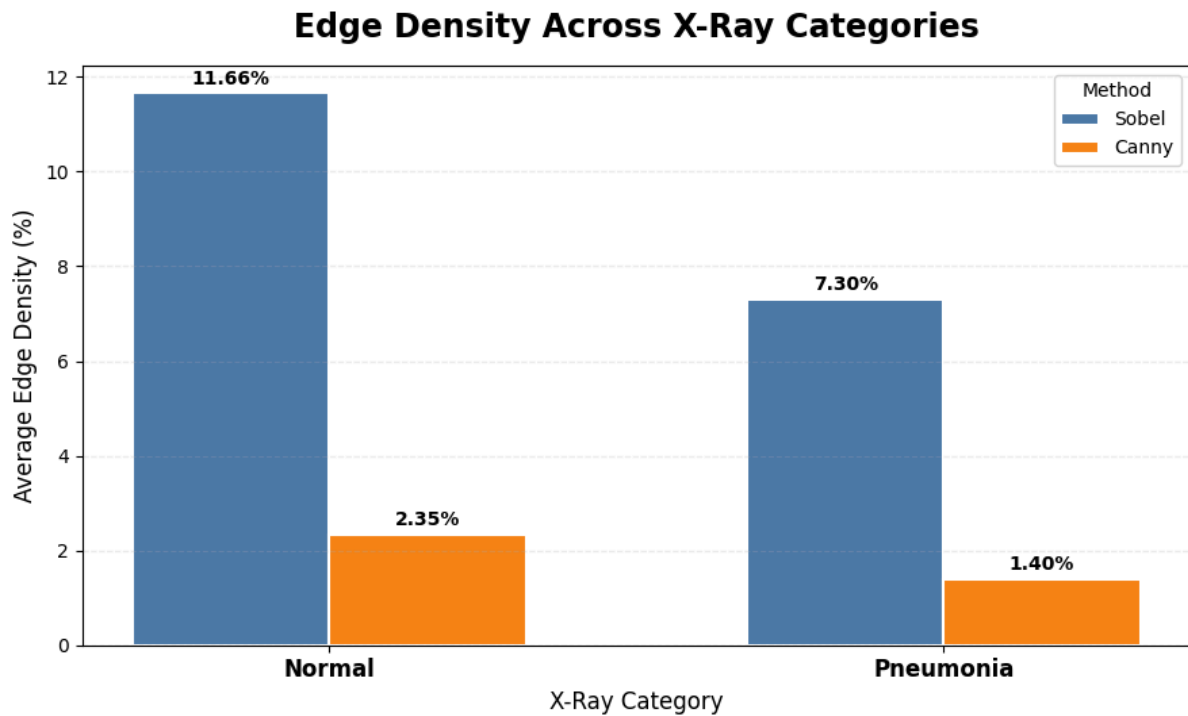


Figure 5: Comparison of average edge density for Normal and Pneumonia X-ray categories using Sobel and Canny.

## Chapter 5: Conclusions and Future Scope

### 5.1 Summary and Key Takeaways

This project presented a comparative study of the Sobel and Canny edge detection techniques for detecting boundaries in medical chest X-ray images. A balanced set of 100 X-ray images, consisting of 50 Normal and 50 Pneumonia images, was used for the experiment. All images were converted to grayscale, resized to 512×512 pixels, and smoothed using a 5×5 Gaussian filter before applying both edge detection methods.

The results showed clear differences between the two techniques. Sobel produced a denser edge representation, with an average edge density of 9.48%, whereas Canny produced an average edge density of 1.87%. Sobel detected broader regions of intensity changes, while Canny generated thinner and more selective boundaries due to its non-maximum suppression and hysteresis thresholding stages. The measured average processing time was 2.873 ms for Sobel and 1.108 ms for Canny under the experimental conditions used in this project.

Overall, the experiment showed that both methods can identify important boundaries in X-ray images, but they produce different types of edge representations. The results also showed that Canny's output is influenced by the selected threshold values, making parameter selection an important factor in the final edge map.

### 5.2 Limitations and Future Scope

The main limitation of this study is that only 100 images were used from the larger dataset. Therefore, the results may not represent all possible variations in chest X-ray images. The comparison was also mainly based on edge density, processing time, and visual observations rather than manually annotated ground-truth boundaries.

Future work can include testing the methods on a larger and more diverse dataset and using adaptive techniques to select suitable thresholds automatically. Other edge detection methods such as Prewitt and Laplacian of Gaussian (LoG) can also be included for a broader comparison. The detected edges could further be used as inputs for image segmentation or machine-learning models to study their usefulness in medical image analysis. Additional evaluation using metrics such as precision, recall, and localization accuracy could provide a more detailed assessment of edge detection performance.

## Chapter 6: References

### 1. Dataset Source & Clinical Data:

- Kaggle Dataset: Mooney, P. T. *Chest X-Ray Images (Pneumonia)*.
- Primary Publication: Kermany, D. S., Goldbaum, M., Cai, W., Valentim, C. C., Liang, H., Baxter, S. L., ... & Zhang, K. (2018). *Identifying Medical Diagnosis and Treatable Diseases by Image-Based Deep Learning*. *Cell*, 172(5), 1122-1131.

### 2. Canny Edge Detection Algorithm:

- Foundational Paper: Canny, J. (1986). *A Computational Approach to Edge Detection*. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, PAMI-8(6), 679 - 698.

### 3. Sobel Operator & Edge Detection:

- Foundational Work: Sobel, I., & Feldman, G. (1968). *A 3x3 Isotropic Gradient Operator for Image Processing*. Talk presented at the Stanford Artificial Intelligence Project.
- Textbook Reference: Gonzalez, R. C., & Woods, R. E. (2018). *Digital Image Processing* (4th ed.). Pearson.